

LIGHTING & SMALL POWER DESIGN REPORT

Project	Duplex
Client	—
Prepared by	AutoDesign
Date	2026-07-10
Standard	BS7671 / EN 12464-1
Supply	230V, 50Hz, 3-phase

1. Lighting Design Calculations

Fixture layout and counts determined by the Lumen (Zonal Cavity) method: $N = (E \times A) / (F \times UF \times MF)$ where E = required illuminance (lux), A = room area (m²), F = fixture flux (lm), UF = utilisation factor, MF = 0.80.

Room	Type	Area (m²)	Req'd Lux	Room Index	UF	Fixture Flux (lm)	N Fixtures	Achieved Lux
kitchen	kitchen	14.6	300.0	0.71	0.44	10000.0	5	1211.8
bedroom1	bedroom	14.6	500.0	0.71	0.45	10000.0	2	486.9
living room	living_room	27.0	500.0	0.97	0.54	10000.0	3	478.6
bedroom	bedroom	12.1	500.0	0.65	0.42	10000.0	2	552.9
bedroom3	bedroom	7.5	500.0	0.51	0.40	10000.0	1	425.7
toilet	bathroom	2.4	150.0	0.30	0.40	10000.0	1	1312.0
hallway	corridor	8.0	100.0	0.43	0.40	10000.0	1	397.6
hallway2	corridor	4.3	100.0	0.38	0.40	10000.0	1	750.8
stairs	other	3.7	150.0	0.31	0.40	10000.0	1	872.3

2. Fixture Schedule

Room	Fixture Type	Qty	Wattage (W)	Total Load (W)
kitchen	FloodLight	5	120.0	600.0
bedroom1	FloodLight	2	120.0	240.0
living room	FloodLight	3	120.0	360.0
bedroom	FloodLight	2	120.0	240.0
bedroom3	FloodLight	1	120.0	120.0
toilet	FloodLight	1	120.0	120.0
hallway	FloodLight	1	120.0	120.0
hallway2	FloodLight	1	120.0	120.0
stairs	FloodLight	1	120.0	120.0
TOTAL		17		

3. Distribution Board Schedule — Ground Floor

DB location: (-8.03 m, -3.51 m) from drawing origin. Incomer: 80 A MCB, 4.0 mm² cable.

Circuit	Description	Pts	Load/pt (W)	Load Factor	Phase Load (W)	V	I (A)	I+25% (A)	MCB (A)	Cable (mm²)
CCT1	LP-R1-01, LP-R1-02,	10	120.0	0.85	1020	230	5.22	6.52	10	1.0
CCT2	LP-Y2-01, LP-Y2-02,	7	120.0	0.85	714	230	3.65	4.57	6	1.0
CCT4	SP-R4-01, SP-R4-02,	6	150	1.0	900	230	4.60	5.75	6	1.0
CCT5	SP-Y5-01, SP-Y5-02,	6	150	1.0	900	230	4.60	5.75	6	1.0
CCT6	SP-B6-01, SP-B6-02,	6	150	1.0	900	230	4.60	5.75	6	1.0
CCT7	SP-R7-01, SP-R7-02,	3	150	1.0	450	230	2.30	2.88	6	1.0
CCT7	SK-R7-04, SK-R7-05,	6	1060	1.0	6360	230	32.53	40.66	50	6.0
CCT8	SK-Y8-01, SK-Y8-02,	5	150	1.0	750	230	3.84	4.80	6	1.0

Normal running load: R = 8730 W Y = 2364 W B = 900 W

Total power: 11994 W at 230 V Full load current: 52.15 A +30% spare: 67.79 A Selected incomer MCB: 80 A

4. Cable Selection Summary

Cable sizes selected per BS7671 Appendix 4, maximum 3% voltage drop (6.9 V at 230 V). Run lengths estimated using Manhattan-distance routing with 1.2× routing factor. All cables: 3-core (L+N+E) PVC/SWA or PVC/PVC twin-and-earth as appropriate.

Circuit	Type	Phase	Cable (mm²)	MCB (A)	V (V)	VP (%)
CCT1	Lighting	R	1.0	10	4.07	1.77%
CCT2	Lighting	Y	1.0	6	2.98	1.30%
CCT4	Socket	R	1.0	6	4.06	1.77%
CCT5	Socket	Y	1.0	6	2.39	1.04%
CCT6	Socket	B	1.0	6	4.47	1.94%
CCT7	Socket	R	1.0	6	1.64	0.71%
CCT7	Socket Key	R	6.0	50	4.52	1.97%
CCT8	Socket Key	Y	1.0	6	3.14	1.37%

5. Design Basis & Assumptions

- Illuminance targets: EN 12464-1 (offices 500 lux, bedrooms 150 lux, kitchens 300 lux, corridors 100 lux).
- Lumen method: maintenance factor MF = 0.80, utilisation factor from simplified zonal cavity table.
- Supply voltage: 230 V, 50 Hz single-phase, 3-phase incomer.
- Cable volt-drop limit: 3% of 230 V = 6.9 V (BS7671 Appendix 12).
- Protective device: MCB to BS EN 60898, next standard rating above 125% of design current.
- Socket-outlet quantities: BS7671 guidance; minimum one per 8 m² for living/kitchen areas.
- DB spare capacity: 30% above calculated full-load current for future load growth.
- Cable routing factor: 1.20 applied to straight-line distances for bends and set-downs.
- This report is produced by automated calculation and must be verified by a qualified engineer.